

Le Hoang Tran

Arlington, VA | hoangtl.cs@gmail.com | (+1) 202-352-0158 | [linkedin](#) | [github](#) | [website](#)

Summary

Backend-focused Software Engineer with strong experience in building large-scale systems. Specialized in high-concurrency, low-latency cloud-native infrastructures and performance optimization. Delivered products that serves millions of users.

Professional Experience

Viettel Group

May 2021 – Jun 2025

Backend Software Engineer

Hanoi, Vietnam

Techstack: Java, Spring Boot, Kafka, Redis, MariaDB, Elasticsearch, MongoDB, MinIO, Camunda, C, JNI, ...

Customer Engagement Platform

2023 – 2025

- Designed and implemented an **in-house marketing system** to support personalized campaigns via push notifications, in-app messaging, and rewards distribution, delivering **100M+ events** to end users daily.
- Optimized user segmentation creation **throughput from 500 to 35,000 QPS**, **reducing p95 latency by 80%** by introducing **Roaring Bitmap-based solution** and multiple caching layers.
- Engineered **workflow automation module** using Camunda, including business process orchestration, rule evaluation, timer events, enabling non-technical operations teams to configure marketing workflows independently.
- Led a team of 4 backend engineers**; driving sprint plannings, system design and code reviews to continuously extend system functionalities, in reaction to meeting user needs and growing demands.
- Monitored system metrics via Grafana** and proactively detect potential areas for performance improvement. Reported to relevant teams, applied multiple **optimization techniques** including configuration tuning, caching strategies, and microservices patterns.

Data Tracking System

2022 – 2023

- Designed and delivered a **high-throughput event tracking system** for mobile and web platforms using Java and Spring Boot, processing **150M+ events/day** and **eliminating third-party licensing costs**.
- Implemented an online tracking rule configuration module, supporting **3,000+ dynamic tracking rules**.
- Reducing data-capture latency by 30%** by introducing decision-tree based engine to optimize payload structure
- Engineered reactive data fetching APIs with **Spring WebFlux** to improve throughput and responsiveness, handling over **150M+ events daily**.

Smart Authentication System

2021 – 2022

- Collaborated in researching and delivering **SmartOTP** solution for **20M+ end users**, eliminating reliance on SMS OTPs, **saving over \$200K annually** and enhance authentication reliability .
- Implemented secure key exchange, OTP generation, and digital signature mechanisms across critical APIs, strengthening authentication and API security.
- Applied **mobile security hardening techniques** (code obfuscation, root/jailbreak/tamper detection) to Android application using native **C/JNI**, passing enterprise-level security audits.

User's Contact System

2024

- Designed a backend module to collect and synchronize user contact data across devices for Viettel Money application, supporting **20M+ end users**.
- Optimized user contacts store using Roaring Bitmap, **reducing storage usage by 70%** and improve contact discovery throughput by 80%.

Education

George Mason University

Aug 2025 – (Expected) Dec 2026

Master in Computer Science | GPA: 4.0/4.0

Fairfax, VA, USA

Hanoi University of Science and Technology

Aug 2017 – May 2021

Bachelor in Information Technology | GPA: top 5% of major

Hanoi, Vietnam

Skills

Languages: Java, Go, JavaScript, C, Python

Backend: Spring Framework, Apache Kafka, Camunda, REST, gRPC, Unit & Integration Testing

Databases: MySQL, MongoDB, Redis/Valkey, Elasticsearch, MinIO, Clickhouse

Infrastructure: AWS, Docker, Kubernetes, CI/CD, Jenkins, SonarQube, Keycloak, ELK, k6

Familiar with: Agile SDLC, Distributed Systems, Design patterns, Cybersecurity, AI-assisted development

Certificates

AWS Certified Solutions Architect – Associate

Sep 2023

Oracle Certified Java SE Programmer

Apr 2023